

Earnings Discontinuity Analysis — Project Guide

Everything needed to explain and defend this project

The result in one paragraph. I tested a famous accounting claim on current SEC data: that companies bend their earnings to avoid reporting a small loss, which should show up as a gap in the distribution of profits at exactly zero. In 16,018 company-years the gap goes in the predicted direction at every bin width but is never statistically significant. Two robustness findings matter more than the headline: the result appears under one scaling choice and not others, and a placebo test on cash flow — which cannot be managed through accounting judgment — shows a *larger, opposite-signed* break. So the honest conclusion is a null, and the interesting contribution is showing how much the classical result depends on choices usually made in passing.

How this is organised. Parts 1–6 are the project in build order. Part 7 covers limitations and phrasing. Part 8 is the code. Part 9 is 26 anticipated questions with answers — the section to read if time is short. A numbers cheat sheet and glossary close it out. Terms are defined the first time they appear, marked like this:

| **Term** — its definition.

PART 1 — BACKGROUND

1. What companies report

Three financial statements, published annually.

Statement	Answers	Covers
Income statement (P&L)	How much did it earn?	A period
Balance sheet	What does it own and owe?	One instant
Cash flow statement	How much cash moved?	A period

Four numbers from them are used here.

| **Net income** — bottom line of the income statement: revenue minus every expense including interest and tax. Positive is a profit, negative a loss. *The number at the centre of this project.*

| **Total assets** — from the balance sheet: everything the company owns that has value (cash, inventory, factories, patents, money owed to it). The standard measure of size.

| **Revenue** — top line of the income statement: money from sales before any costs. A company can have huge revenue and still lose money.

| **Cash flow from operations (CFO)** — cash the actual business generated, excluding money raised from investors or spent on acquisitions.

One structural fact matters later. A balance sheet always balances:

Assets = Liabilities + Shareholders' equity

| **Liabilities** — what the company owes: loans, bonds, unpaid bills.

| **Shareholders' equity** (book equity) — assets minus liabilities; what would be left for owners after paying every debt.

Profit flows into equity. Unpaid-out profit lands in *retained earnings*, inside equity, inside assets:

This year's profit → retained earnings → equity → total assets

That chain is why the *timing* of the assets figure is the most important measurement decision in the project (§11).

Fiscal year — a company's own 12-month reporting period. Need not be the calendar year: Apple's ends in September, many retailers' in January. I always compare a company's fiscal year to its own prior fiscal year.

2. Accruals, and why cash flow is different

Accrual accounting — record revenue when *earned* and expenses when *incurred*, regardless of when cash moves. Required for public companies.

Example: a company delivers software in December 2024 and is paid in February 2025. Cash accounting counts the sale in 2025; accrual accounting counts it in 2024, when the work happened. Accrual is more informative — but it requires estimates, and estimates require judgment.

Accrual — an entry recording revenue or expense before the cash moves. The gap between profit and cash is made of accruals.

Four places judgment legitimately enters:

- **Bad debt allowance** — some credit customers will never pay. Is the provision 2% or 3%?
- **Warranty reserve** — estimating today what future repairs will cost.
- **Depreciation life** — a 10m machine over 8 years is 1.25m/year; over 12 years it is 833k. A 417k annual difference from one judgment.
- **Write-off timing** — when the evidence becomes conclusive enough can fall either side of a year end.

None of these are cheating. Auditors sign off on ranges, not single correct answers.

The contrast that makes this project work:

	Net income	Cash flow from operations
Contains accruals	Yes, substantially	No
Movable by revising an estimate	Yes	No
Movable by real operational action	Yes	Yes, at real cost

You cannot change the bank balance by revising a warranty estimate. **This asymmetry is the entire basis of the placebo test in §20.**

3. Earnings management, and why zero

Earnings management — using discretion available *within* accounting rules to steer reported results. Legal. Not fraud.

	Earnings management	Fraud
Within the rules	Yes	No
Example	3% bad-debt provision rather than 2%	Recording a sale that never happened
Legal	Yes	No

Zero has no economic meaning — a company earning 1 is not healthier than one losing 1. But it has enormous contractual and psychological meaning:

- **Headlines.** "Posts loss" is a story; "posts small profit" is not.
- **Debt covenants.** Loan agreements often require profitability; breaching one can let a lender demand immediate repayment.
- **Bonuses.** Many schemes pay out only if the company is profitable.

- **Loss aversion.** People treat losses as worse than equivalent gains.

The testable chain: zero matters to managers → accruals give them discretion → some may use it to cross zero → if enough do, it shows in the shape of the distribution across thousands of companies. You cannot see one company's decision, but you can ask whether the aggregate pattern looks natural.

4. Distributions and bins

Distribution — the pattern of how often each value occurs across a group.

Histogram — a chart of a distribution: split the range into equal **bins**, count what falls in each, draw bars.

Bin — one interval, written `[left, right)`. The left edge belongs to the bin, the right edge to the next one.

Consequence here: a company reporting *exactly* zero lands in the bin **above** zero. Breaking even is not a loss.

Why we expect smoothness. Business conditions vary continuously, so the count of companies earning 0.1% should be close to the count earning 0.2%. A sudden break demands explanation, because economics does not jump.

Bin width is a real decision, not a display setting.

- *Too wide* — the affected companies get mixed with thousands of unaffected ones and the signal is averaged away.
- *Too narrow* — each bin holds a handful of companies, so 3 versus 7 looks dramatic when it is just noise.

There is no single correct width, so **I run every test at three: 0.0025, 0.005, 0.01, and always report all three.**

That removes the question "did you pick the flattering one?" — and it turned out to be diagnostic, because in §20 an effect *grows* with bin width, which is itself evidence about its cause.

Zero must sit on a bin edge. A bin running -0.25% to $+0.25\%$ would contain both the deficit below zero and the surplus above, which cancel inside the same bar — destroying the signal before measuring it. Every bin edge in my code is an integer multiple of the bin width, so zero is always a boundary. A unit test enforces this, including at awkward widths like 0.003 where the window does not divide evenly.

PART 2 — THE QUESTION AND THE DEBATE

5. The hypothesis

Line every company up by profit as a fraction of its size and count them into bins.

- **If nobody manages anything**, the distribution passes smoothly through zero.
- **If some managers push small losses into small profits**, there is a *deficit* just below zero and a *surplus* just above.

Notch (discontinuity) — a sharp break in an otherwise smooth distribution. Here, the dip-then-bump at exactly zero.

Null hypothesis (H₀) — the "nothing unusual" default: the distribution is smooth and any apparent notch is random variation.

Alternative (H₁) — the bin below zero is genuinely thinner than its neighbours predict, and the bin above genuinely thicker.

Two commitments this forces, both of which I kept:

1. **Report both sides, every time.** The prediction is a deficit below *and* a surplus above. Reporting only the better-looking side would overstate the finding. In my results the two sides behave quite differently, so this matters.
2. **The direction is predicted in advance.** A large effect pointing the *wrong* way is not support. §20 turns on exactly that distinction.

6. The two papers

Burgstahler & Dichev (1997), *Journal of Accounting and Economics* 24(1), 99–126. Examined US companies 1976–1994, found the notch, read it as earnings management to avoid losses. One of the most-cited results in empirical accounting. They also tested a second threshold — zero change in earnings year-over-year — which I did not test (§26).

Durtschi & Easton (2005), *Journal of Accounting Research* 43(4), 557–592. The critique, and the reason this project is more than a replication.

Their argument starts from something unavoidable: you cannot compare raw profit across companies, because \$1m means different things to a corner shop and to Apple. Every study therefore **divides profit by something**. They showed that the act of dividing, plus the choice of which companies end up in the sample, **can manufacture a notch with no manager involved**.

One mechanism, simplified. Small companies are less profitable and more variable than large ones. Dividing a modest profit by a huge asset base gives a ratio near zero; dividing by a tiny asset base gives a ratio far from zero. So the region near zero fills with large stable companies and the tails with small volatile ones. **The sample near zero is a different population from the sample further out**, and the join between two populations with different shapes can produce a kink. Sample selection compounds it: requiring complete data removes companies non-randomly, and if the removal rate differs either side of zero, the shape shifts.

This dictates the whole design. The filters in §9 exist because selection is half the critique. The four robustness checks in §17–20 each target a specific piece of it.

Both citations were verified against the publishers, not quoted from memory. A longer 2009 follow-up by the same authors exists; I mention it in the README but do not rely on it, having not verified its pages.

PART 3 — DATA AND SAMPLE

7. Source and files

SEC — the US regulator of public companies. All filings go to its public **EDGAR** database, free to anyone.

10-K — the audited annual report. **10-Q** — the unaudited quarterly report.

XBRL — a tagging standard attaching machine-readable labels to every number in a filing. Phased in from ~2009. It is what makes this project possible without a data licence — and why my data cannot reach earlier periods (§22).

SEC Financial Statement Data Sets — free quarterly ZIPs flattening every XBRL filing into tab-separated text, at sec.gov/files/dera/data/financial-statement-data-sets/{YYYY}q{Q}.zip.

I used **37 quarterly archives, 2017 Q1 – 2026 Q1**. Chose this over a commercial database because it is free, primary (no vendor has made undocumented decisions), and fully reproducible by anyone. The cost is coverage starting in 2009.

Each archive holds two files I use:

- `sub.txt` — one row per filing: accession number, **CIK** (the SEC's permanent company ID, stable across renames — the key I join on), name, SIC industry code, period end, form type, filing date.
- `num.txt` — one row per reported number: filing, XBRL tag, date, `qtrs`, unit, `segments`, `coreg`, value. ~550 MB uncompressed per quarter.

Three fields need care:

- `qtrs = 0` is a point-in-time balance-sheet value; `qtrs = 4` is a full year. Getting this wrong would mean treating one quarter of profit as a year's.
- `segments` and `coreg` are non-empty for segment or subsidiary breakdowns. **I keep only rows where both are blank**, so every figure is a company-level consolidated total. Without this I would be mixing totals with regional

subtotals.

- `ddate` is rounded to month end, and 52/53-week fiscal years drift, so dates are matched by window rather than exactly (§8).

Why I parsed by hand rather than with `read_csv`. Three reasons: 20 GB of text total, and loading a quarter to then filter to six tags wastes most of the work; the free-text `footnote` field contains quote characters that break naive CSV parsing silently; and the usual remedy — skipping malformed lines — would drop data without counting it, which is backwards in a project whose credibility rests on counting every exclusion. Splitting on tabs and filtering before materialising anything keeps memory flat. **Result: 113,203,537 rows scanned, zero malformed.**

Tags extracted. Several concepts have multiple tags in common use, so I take the first available in a stated priority order:

Concept	Tags, in priority order
Total assets	Assets
Net income	NetIncomeLoss, ProfitLoss
Revenue	Revenues, RevenueFromContractWithCustomerExcludingAssessedTax, + 3 older variants
Cash flow from ops	NetCashProvidedByUsedInOperatingActivities, + continuing-ops variant
Equity	StockholdersEquity, + variant including non-controlling interests

Insisting on one tag would lose every company using a newer concept. **Revenue is the messiest** — five plausible tags, inconsistently used — which is why the revenue-scaled sample in §17 is a different set of companies, a caveat I raise there.

8. Building the panel

Panel — one row per company per year. Each row is a **firm-year**, the unit of analysis. "16,018 observations" means 16,018 company-year combinations, not 16,018 companies.

Why 10-Ks only — the key data decision. To compute my measure I need this year's profit and the assets the company *started* the year with. The obvious approach is to fetch two filings and join them, which risks missing years, changed year-ends, and silent mismatches.

But accounting rules require a 10-K to show the **comparative balance sheet**: last year's figures printed beside this year's, in the same audited document, tagged in the same XBRL submission. So one filing gives me all three of: net income for the year, end-of-year assets, and **beginning-of-year assets**. No join, no mismatch risk. Quarterly filings do not provide this, which is why they were excluded despite the volume lost.

Date matching. Current year = within ± 20 days of period end. Prior year = 330–400 days before period end. The window is generous enough to absorb a 53-week year and month-end rounding, tight enough not to grab a figure from two years ago.

Deduplication. Amended and transition-period filings can duplicate a firm-year; I keep the most recent period end. This removed only 4 rows of 31,244 — a safety net that caught almost nothing, which is the good outcome.

Caching. The cleaned panel is saved as **parquet** (a compressed, typed, column-oriented format) and *committed to the repository*. Raw archives are gitignored. So anyone can clone and reproduce every number without downloading 2 GB, and the notebooks never touch the SEC.

9. The filters

Why filter: a bank's balance sheet is not comparable to a manufacturer's, missing data cannot be measured, and a company with \$50k of assets produces ratios that distort everything.

Why filtering is dangerous: every filter is a chance to p-hack.

p-hacking — trying many analytical choices and keeping the most favourable. Inflates apparent significance and causes results that fail to replicate.

Three protections: **(1)** filters were fixed before any result existed; **(2)** every exclusion is *counted*, not described — `apply_filters()` returns the clean panel *and* a drop log, and a test verifies each step's "before" equals the previous "after"; **(3)** the one threshold I chose myself is stress-tested at six values (§19).

Step	Dropped	Remaining	Firms
0. All 10-K firm-years	—	57,407	10,304
1. Fiscal years 2016–2025 only	141	57,266	10,276
2. Drop missing industry code	550	56,716	10,078
3. Drop financial firms (SIC 6000–6999)	14,590	42,126	7,251
4. Drop utilities (SIC 4900–4949)	1,404	40,722	7,034
5. Drop missing net income	703	40,019	6,979
6. Drop missing / non-positive lagged assets	764	39,255	6,882
7. Size floor: lagged assets $\geq$ \$10m	8,011	31,244	5,366
8. One filing per firm-year	4	31,240	5,366

1. Fiscal-year range. My download window only fully covers 2016–2025. Outside it are delinquent filers catching up — fiscal 2014 appears exactly once. One company is not a cross-section.

2. Missing SIC. Needed for the next two filters.

SIC code — a four-digit industry classification. 2834 is pharmaceuticals, 6021 is commercial banks.

3. Financial firms — the largest cut, a quarter of the data. "Total assets" means something structurally different for a bank: its assets are its *loan book*, not the factories and inventory it uses to make money. A bank might hold 100 of assets per 1 of annual profit; a software company \$2. Both are healthy, but the ratio is not comparable. Critically, per \$6, mixing populations with different natural scales is exactly what can manufacture a shape artifact — so including banks would mean any kink was partly my own construction. Standard practice, and the count is in the log.

4. Utilities. Rate regulation means a regulator effectively sets the return, making earnings mechanically smooth near a target for reasons unrelated to discretion. Conventional but genuinely optional; it is a flag in my code and logged either way.

5. Missing net income. No numerator, no observation.

6. Missing or non-positive lagged assets. Zero assets gives infinity. *Negative* assets flip the sign — a company with a 5m loss and –10m of assets computes to +0.5, landing on the profit side of zero. In a study about which side of zero companies fall on, that is not a small problem.

7. Size floor of 10m. Very small denominators produce extreme ratios (50k of assets, \$30k loss $\rightarrow$ –60%), and these are precisely the companies Durtschi and Easton identify as the artifact source. \$10m is small enough to keep genuine small-caps, large enough to remove shells. **This is the only threshold I chose myself, so §19 re-runs everything at six floors from 0 to 500m.**

8. One filing per firm-year. §8.

10. The window, and pooling

The analysis window. At test time I restrict to scaled profit within $\pm 10\%$: 16,018 firm-years inside, 15,222 outside.

This is *not* a data-quality filter — it states **what is being studied**. A company on track for a 40% return is not weighing whether to cross into profit. It is reported separately in the drop log for that reason. It was fixed in advance and is conventional. It also matters less than it looks: the neighbour test depends only on the *three bins around zero*, which are identical regardless of where the window ends.

Pooling. 31,240 firm-years from 5,366 companies is **5.82 years per company**.

Pooling — treating every firm-year as a separate observation. Violates independence, because a company profitable last year is likely profitable this year.

Consequence: **my p-values are somewhat too small** — results look more certain than they are. I pooled anyway because it is standard in this literature (both framing papers do it), because keeping one year per company would cut the sample to ~5,400 and destroy the power to detect anything, and because **the bias runs toward significance while my finding is a null** — correcting for it would make the null *more* null, not less.

I raise this before being asked, because stating a limitation you could have concealed is what makes the ones you did not state credible.

PART 4 — MEASUREMENT

11. Scaling, and the lagged denominator

Why divide at all. Two companies each report 1m of profit: a restaurant chain with 3m of assets, and Walmart with \$250bn. Same number, opposite meanings. Raw profit cannot be compared, so every study scales.

Scaling — dividing by a measure of size so companies of different sizes are comparable.

ROA (return on assets) — net income $\div$ total assets. My main measure.

Which total assets — the single most important decision in the project.

The intuitive answer is end-of-year, because that is the headline figure. **That is wrong.** Recall from §1: profit $\rightarrow$ retained earnings $\rightarrow$ equity $\rightarrow$ assets. **End-of-year assets already contain this year's profit.** Dividing by them puts the measured quantity into its own denominator.

Watch what that does. A good year means high profit, which inflates the denominator, which shrinks the ratio. A bad year shrinks assets, which inflates the (negative) ratio. **Both directions get pulled toward the middle — the distribution is compressed toward zero.** That is catastrophic for a test about a sharp feature *at* zero: it acts as a smoothing filter applied exactly where I am looking, and I could not tell a genuinely absent notch from one I erased myself.

The fix: use the assets the company *started* the year with.

$$\text{ROA} = \text{net income during year } t \div \text{total assets at end of year } t-1$$

Lagged — taken from the previous period. Lagged assets = last year's closing balance sheet = this year's opening one.

That figure was finalised before this year's profit existed, so it cannot contain it. It also asks a cleaner question: how much profit came from the resources available at the start.

How much it matters, measured rather than assumed:

Check	Result
Firm-years where beginning = ending assets	0.02%
Median absolute difference in resulting ROA	0.0093

0.0093 is nearly **two bins wide** at my main width of 0.005. Not a rounding difference — it would move many companies across the bins the entire test depends on.

Three protections. The comparative balance sheet (§8) puts the right number at hand. A unit test uses a synthetic company with 100 of assets in year 1, 400 in year 2, earning \$20 in year 2, and asserts ROA is 0.20 not 0.05 — so switching to the end-of-year figure fails immediately. Companion tests confirm a first year has no lag, a *gap* year does not silently become the lag, and lags never cross between companies.

12. The other scalars

Scalar	Formula	Weakness
Revenue	net income ÷ revenue (profit margin)	Messiest XBRL concept; 28,332 firm-years have it vs 31,240 with assets, so a different company set
Book equity	net income ÷ lagged equity (ROE)	Equity can be negative, flipping the sign — smallest sample, 7,177 in window
Market value of equity	net income ÷ (price × shares)	Not available — SEC filing data contains no share prices

Market value is arguably the *best* scalar, because it is the only size measure not produced by the same accounting system whose output I am testing. But the SEC files have no prices. I used book equity on the main sample, tested market value on my 300-ticker prototype and got only 237 in-window observations — too few to inform anything — and **reported it as a documented gap** rather than quietly dropping the check.

The placebo variable:

cash flow from operations ÷ total assets at the start of the year

Note the denominator is **identical** to the main measure. Only the numerator changes. That is deliberate: if the scaling manufactures a notch, it will manufacture one here too.

PART 5 — THE STATISTICS

13. Hypothesis testing

I observe 459 companies in the bin below zero and expected about 480. Is 21 fewer meaningful? You cannot tell by looking, because counts always vary.

Sampling variation — the wobble in any measurement from a limited number of observations, present even when nothing underneath is changing.

The procedure assumes the null is true and rejects it only if the data would be very unlikely under it — like assuming innocence and requiring strong evidence to convict. The asymmetry is intentional: it makes accidental "discoveries" hard.

p-value — the probability of a result at least this extreme *if the null were true*. $p = 0.43$ means a gap this big happens 43% of the time by chance. $p = 0.001$ means once in a thousand.

Significance level — the conventional threshold, 0.05.

What a p-value is not: the probability the hypothesis is true, or that you are wrong. It is only a statement about compatibility with the null. This matters here, where two of my p-values land at 0.051 and 0.057 — just the wrong side of a line that is itself arbitrary.

Type I error (false positive) — concluding something is happening when nothing is. **Type II error (false negative)** — missing a real effect, usually from too little data.

Both are live. My null could be a false negative if the effect is real and small. My one significant result could be a false positive, since running many specifications raises the chance one crosses the line by luck.

Significance is not importance. With a large enough sample, a trivial effect becomes detectable. Even the significant revenue-scaled notch is a shortfall of ~69 companies out of 13,452.

14. The test

The idea: predict each bin's count from its two neighbours.

$$\text{expected count in bin } i = (\text{count in bin } i-1 + \text{count in bin } i+1) \div 2$$

This makes one minimal assumption — that over a short stretch the distribution is roughly straight — rather than requiring me to choose a functional form for the whole curve and risk inventing the answer. It is a *local* assumption, which is its strength and, per §20, also its weakness.

Worked through with the real numbers, bin width 0.005, the bin just below zero:

Bin	ROA range	Count
The one before	-0.010 to -0.005	409
Being tested	-0.005 to 0	459
The one after	0 to +0.005	550

$$\begin{aligned}\text{expected} &= (409 + 550) \div 2 = 479.5 \\ \text{gap} &= 459 - 479.5 = -20.5\end{aligned}$$

Negative — the predicted direction. Is -20.5 a lot?

Standard error — the typical amount a statistic varies from randomness alone; the yardstick a gap is measured against.

Where it comes from. Each company either lands in this bin or does not — a yes/no outcome across N companies.

Binomial distribution — the count of successes in N independent yes/no trials, with variance $N \cdot p \cdot (1-p)$.

Variance — the square of the standard deviation. Variances of independent quantities add, which is why the derivation works in variance units and takes a square root at the end.

The bin's own count contributes $N \cdot p_i \cdot (1-p_i)$. The prediction is built from two random neighbour counts, so it wobbles too, contributing $N \cdot (p_{i-1} + p_{i+1}) \cdot (1-p_{i-1} - p_{i+1})$ — but I use their *average*, and halving a quantity quarters its variance since $(1/2)^2 = 1/4$. Adding:

$$\begin{aligned}\text{variance} &= N \cdot p_i \cdot (1-p_i) + \frac{1}{4} \cdot N \cdot (p_{i-1} + p_{i+1}) \cdot (1-p_{i-1} - p_{i+1}) \\ \text{standard error} &= \sqrt{\text{variance}} = 25.9\end{aligned}$$

z-score — how many standard errors an observation sits from its expected value.

$$z = -20.5 \div 25.9 = -0.79$$

z	Reading	≈ p
under 1	Completely ordinary	above 0.32
1 – 1.96	Notable, not significant	0.05 – 0.32
1.96	The 5% threshold	0.05
over 2.58	Strong	under 0.01
over 3.29	Very strong	under 0.001

z = −0.79; right direction, unremarkable size. The same calculation for the bin above zero gives observed 550, expected 496, **z = +1.95** — close to the threshold. The two sides behave differently, which is precisely why I report both.

15. A second test, and the three guardrails

The held-out fit. The neighbour method assumes local straightness, which is doing real work. So a second test gets its expectation differently: fit a degree-4 polynomial across the window **with the two zero-adjacent bins removed**, then test those two bins against the fit.

Goodness-of-fit test — compares observed counts to a model's predictions across many bins at once.

Chi-square statistic — sums $(\text{observed} - \text{expected})^2 \div \text{expected}$ across bins. Large values mean poor fit. Its **degrees of freedom** determine how large counts as surprising.

Because the curve never saw the tested bins, this is a genuine out-of-sample comparison with 2 interpretable degrees of freedom. It rejects smoothness at all three widths ($p = 0.0009, 0.0052, 0.0044$) — **but decomposing it, almost all of the rejection comes from the surplus above zero, not a deficit below.** Two methods, different routes, compatible conclusion: whatever is happening near zero is on the profit side.

The three guardrails, all fixed in code before any result existed:

- 1. Zero on a bin edge.** §4. Edges are generated as integer multiples of the width rather than by stepping from the left, because stepping accumulates floating-point error and after forty steps "zero" might be 0.0000000001. Tested at all three widths plus awkward ones, and for the exact-zero and −1e-9 edge cases.
- 2. Three bin widths, all reported.** §4. Became a diagnostic tool in §20.
- 3. Both sides of zero, always.** §5. My two sides behave differently; reporting only the above-zero side would have made this look like a positive result.

All three are the same principle: **make the decision before you can see which answer it produces.**

PART 6 — RESULTS

16. The main result

Net income ÷ beginning-of-year assets. N = 16,018 firm-years, 3,582 companies, FY2016–2025.

Bin width	Below zero (obs / exp)	z	p	Above zero (obs / exp)	z	p
0.0025	246 / 252.0	−0.31	0.753	291 / 252.5	+1.91	0.057
0.005	459 / 479.5	−0.79	0.429	550 / 496.0	+1.95	0.051
0.01	868 / 898.5	−0.87	0.383	1,083 / 1,017.0	+1.73	0.083

Encouraging: the sign is correct at every width — negative below, positive above. Pure noise would scatter the signs.

Not encouraging: the deficit below zero is nowhere near significant ($p = 0.75, 0.43, 0.38$ — gaps that size occur 40–75% of the time by chance). The surplus above zero sits *exactly* on the threshold (0.057, 0.051, 0.083).

That is the most uncomfortable place for a result to land. A result at $p = 0.051$ and one at 0.049 are the same evidence; only one gets a label. Treating 0.051 as a discovery is exactly the reasoning that fills literatures with findings that do not replicate. **So I call it marginal and let the robustness checks decide how much weight it carries.**

Summary: weak, directionally consistent, not significant. A less thorough project would have stopped here and said "there is a hint of something on the profit side."

Robustness check — re-running an analysis under different reasonable choices to see whether the conclusion survives. If a finding holds only under one exact combination, it is a property of your choices rather than the world.

17. Robustness 1: does the divisor drive it?

Same companies, same years, same code, same widths. **Only the denominator changes** — so any difference cannot be a period, filter, or method effect.

Profit divided by	N	z below	p	z above	p
Beginning-of-year assets	16,018	-0.79	0.43	+1.95	0.051
Revenue	13,452	-2.97	0.003	+4.62	<0.001
Beginning-of-year book equity	7,177	-1.20	0.23	+1.61	0.11

Revenue holds across widths, ruling out a single-binning fluke: z below = -3.31 / -2.97 / -3.25, all $p \leq 0.003$.

Same firms. Same net income. Three divisors. Three different answers. This is the Durtschi–Easton argument arriving as an empirical fact in 2016–2025 data. The presence of a detectable notch is partly a property of the scaling choice. It also means a researcher who ran only the revenue specification would report a strong finding that does not hold under two equally reasonable alternatives.

The caveat I volunteer. Two explanations, and **I cannot separate them.** (A) Revenue genuinely behaves differently as a divisor. (B) The revenue sample is a different set of companies — 13,452 versus 16,018, because revenue is the messiest XBRL concept (\$7) and companies without clean tagging drop out. Both are consistent with the critique, which names scaling *and* selection, but they are different mechanisms. Separating them needs a matched sample restricted to companies with all three denominators. That is the first thing I would run with more time.

18. Robustness 2: is it a small-company artifact?

Durtschi and Easton make a falsifiable prediction: **if the notch is an artifact, it should be strongest among small firms.**

Tercile — one of three equal groups formed by sorting.

Tercile	Median assets	N	z below	z above
Small	\$196m	5,340	-0.25	+1.38
Medium	\$1.5bn	5,339	-0.26	+1.23
Large	\$9.6bn	5,339	-0.91	+0.73

No concentration in small firms — if anything the mild deficit is largest among the *biggest*. That cuts against this specific mechanism.

Statistical power — the probability of detecting an effect that is genuinely there. Smaller samples have less.

Honest limitation: splitting into thirds cuts power, and with nothing significant in any tercile this is *weak* evidence against the small-firm story, not strong. I would not lean on it.

19. Robustness 3: does my own threshold matter?

The \$10m floor is the only parameter I chose myself, so it is the obvious place to suspect tuning. The answer is not an argument but a table — produced by re-running the **entire filter chain and test** at six floors.

Floor	Panel N	Window N	z below	z above
\$0	39,242	16,480	−0.72	+1.93
\$1m	34,941	16,386	−0.82	+1.95
\$10m (headline)	31,240	16,018	−0.79	+1.95
\$50m	26,315	14,929	−1.09	+2.08
\$100m	23,782	14,193	−0.74	+1.85
\$500m	16,572	11,384	−0.75	+1.46

Flat. Across a fifty-fold range, z below stays between −0.72 and −1.09. The largest floor removes more than half the panel and moves the statistic ~0.3 standard errors. My chosen threshold is doing no work.

One cell worth flagging rather than promoting: at the \$50m floor, z above reaches +2.08, crossing 1.96. If I were hunting significance I could report that specification. I will not — it is one cell of eighteen, and picking it is precisely the p-hacking described in §9. The honest reading is that the above-zero statistic hovers near 1.9 and occasionally tips over by chance. **Reporting the full table is what makes that reading possible.**

20. Robustness 4: the cash-flow placebo

The most important result in the project.

Placebo test — running the same analysis on something the hypothesis says should show *no* effect. If an effect appears anyway, the method is suspect. (From medicine: if the sugar pill works too, the improvement is not from the drug.)

Per §2, cash flow contains no accruals, so it cannot be moved by revising an estimate. Run the identical test with the identical denominator, changing only the numerator. Two expected outcomes:

Outcome	Meaning
Notch in net income, none in cash flow	Effect is specific to accruals — supports management
Notch in both, same direction	Something common produces it — suspect the scaling

Neither happened.

Measure	Width	z below	p	z above	p
Net income	0.0025	−0.31	0.75	+1.91	0.057
Net income	0.005	−0.79	0.43	+1.95	0.051
Net income	0.01	−0.87	0.38	+1.73	0.083
Cash flow	0.0025	+0.45	0.65	−0.58	0.56
Cash flow	0.005	+2.04	0.042	−1.79	0.074
Cash flow	0.01	+3.55	<0.001	−3.59	<0.001

Cash flow shows a break **larger than net income's and pointing the opposite way** — a surplus just below zero, a deficit just above. At the widest width, highly significant in both directions.

This is not the shared-artifact result. A common scaling artifact would push both the same way, since the denominator is identical. The signs are opposite.

What it actually means, which is worse. Cash flow cannot have been managed into an opposite-signed break at zero — there is no accrual mechanism, and no reason to target zero cash flow. Yet the test reports one, significantly. **So the test's core assumption of local smoothness fails — and it is the same machinery, same window, same denominator, that produced the marginal z-scores for net income.** The placebo does not merely fail to confirm; it undermines confidence in the instrument.

The clue in the bin-width pattern. The cash-flow effect grows monotonically: $+0.45 \rightarrow +2.04 \rightarrow +3.55$. A genuine knife-edge discontinuity would behave the *opposite* way, weakening as wider bins dilute a narrow feature with unaffected neighbours. Something that strengthens with width is consistent with **curvature** being mistaken for a break: the neighbour method assumes straightness, and wider bins reach further into regions where the curve bends more, so the miss grows. **This is the payoff from guardrail 2** — with one bin width I would have had a number and no way to interpret it.

What I cannot claim. I have not *proven* the net income result spurious — I have shown the method produces false positives on a related measure, which is grounds for caution, not a demonstration. I have not identified precisely what creates the cash-flow break; curvature is my best explanation and the width pattern supports it, but I have not modelled the density to confirm. What I can say is narrower: **the smoothness null is not safe on this data, and any result resting on it deserves less weight than its p-value suggests.**

21. Conclusion

Four findings. (1) The main test is weak and directionally consistent — right sign at every width, deficit below never significant, surplus above on the 0.05 line. (2) The result depends on the divisor: revenue gives a clean significant notch, assets and equity give nothing. (3) It is not a small-company artifact — no tercile concentration, flat across size floors. (4) The placebo breaks the test.

The conclusion: I did not find statistically significant evidence that companies bend earnings to avoid reporting a loss, in recent SEC filing data.

Why this is a contribution rather than a failure. Read as a *measurement study*, the project answers a real question: how much does the classical result depend on choices usually made in passing? A great deal. The scalar changes whether you find anything; the statistical null fails on a control variable. Both are things anyone building on this literature would want to know, and neither is visible if you run one specification.

There is also a structural point: I designed the project so a null would still be complete — the plan was always "here is the distribution near zero, here is how it responds to bin width, scaling and filters, here is what would have to be true for the classical result to hold." That produces finished work either way. A design that is only interesting if the answer is yes creates quiet pressure to find a yes.

What would have changed my mind — worth stating, because a hypothesis you cannot imagine rejecting is not being tested. All three of: a deficit below zero significant at all three widths, surviving all three denominators, and **absent** from the placebo. I got none of the three. I would also have wanted the effect not to grow with bin width.

PART 7 — LIMITATIONS AND PHRASING

22. Limitations

- **Observations are not independent.** 5.82 years per company; p-values somewhat too small (§10). The bias runs toward significance and my finding is a null, so correcting would make it more null.
- **The period is 2016–2025 only.** The original used 1976–1994. If **Sarbanes-Oxley** (2002 legislation after Enron, requiring executives to personally certify financial statements and strengthening auditor independence) made earnings management riskier, the notch may genuinely have faded — in which case **a null today is compatible**

with the 1997 paper, not a refutation. I cannot test this: SEC structured data begins in 2009, so I cannot reach the pre-SOX period at all. It would need a different data source.

- **The null hypothesis is the weak link.** My own placebo showed neighbour-averaging failing. A better design would fit a flexible model to the whole density and test a break against it — which would also give a *confidence interval* on the notch size rather than only a test against zero. My held-out polynomial test is a step toward that, but only a step.
- **Survivorship.** XBRL-era filers only; companies that went bankrupt or deregistered before 2017 never appear. Loss-makers are likelier to disappear, so the region just below zero is probably affected more than above — which would push *toward* an apparent notch. I cannot measure the size of this.
- **Revenue tagging is inconsistent** (§17), so I cannot separate scalar effects from sample effects.
- **Market value of equity untested at scale** (§12).
- **10-K only; amendments excluded** rather than merged. An amendment sometimes corrects a material number, so in principle the amended figure is better; merging correctly is fiddly and I judged the gain small against the risk. A simplification, stated as one.

What I am *not* limited by: sample size is adequate (16,018 is comparable to the original paper's); the data is primary, not a vendor's cleaned version; every exclusion is counted; and the result is fully reproducible from the committed panel.

23. Phrasing discipline

Allowed: "consistent with earnings management." The test detects an anomaly in the shape of a distribution. It never sees a manager, a decision, or a journal entry. If a notch exists, management is *one* explanation among several — including the scaling effects of §6, the survivorship of §22, and whatever produces the cash-flow break in §20, which is definitely not management.

Banned: "proves managers manipulate earnings." Three errors in five words. *Proves* — statistics quantifies compatibility, it does not prove. *Manipulate* — implies wrongdoing, but earnings management is legal discretion inside the rules. *Managers* — attributes an aggregate pattern to individual intent.

The same restraint applies to my own null. My result is "I did not find evidence," not "there is no effect." Absence of evidence is not evidence of absence, particularly with a marginal result and acknowledged limitations.

Written out properly: *In 16,018 firm-years of SEC data from FY2016–2025, net income scaled by beginning-of-year assets shows a deficit immediately below zero and a surplus immediately above at all three bin widths. Neither is statistically significant under the headline specification. The pattern is consistent with earnings management to avoid losses, and also with several alternatives the robustness checks cannot rule out. A cash-flow placebo shows the method producing significant breaks in a measure that cannot plausibly be managed, which reduces the weight the headline result can carry.*

PART 8 — THE CODE

24. How it is built

Python 3.13 (pandas, numpy, scipy, matplotlib, pyarrow). ~3,200 lines across `src/` and `tests/`; 35 unit tests. Repo: github.com/riddhi-shedge/earnings-discontinuity.

```

src/    sec_loader    download and reduce 37 quarterly archives
      panel          filters, drop log, scaled variables
      binning        bin edges – enforces zero-on-a-boundary
      discontinuity   neighbour test, z-statistics, chi-square
      plotting        publication figures
      build_panel     one command to rebuild the cached panel
notebooks/ 01_explore · 02_main_test · 03_robustness
data/  panel.parquet (committed), drop_log.csv, result tables
tests/ 35 unit tests

```

Loading, filtering, binning and testing are separated so binning can be unit-tested without touching SEC files and a test re-run does not re-parse 20 GB.

The drop log is a first-class object. `apply_filters()` returns `(clean_panel, drop_log)` – a table with one row per filter recording name, written rationale, rows before, dropped, after, and firms remaining. Three consequences: the README table is *generated* from it so documentation cannot drift from code; a test verifies each step's "before" equals the previous "after"; and adding a filter without a rationale is awkward because the signature requires one. **The structure enforces the discipline rather than relying on memory.**

What the 35 tests protect – tests that only confirm code runs are not worth much:

- **Zero-on-a-boundary**, across all three widths plus awkward ones (0.003, 0.0001) where the window does not divide evenly; that exactly one bin is flagged each side; that exactly 0.0 lands above and $-1e-9$ lands below.
- **The lagged denominator** – the 100/400/\$20 synthetic case asserting $ROA = 0.20$ not 0.05, plus no-lag-on-first-year, gap-years, and no cross-company leakage.
- **Drop-log arithmetic.**
- **Does the test actually work?** I generate 200,000 smooth random observations, move half the bin below zero into the bin above – injecting a notch of *known* size – and assert detection in both directions ($z < -5$, $z > +5$). **And the mirror:** on smooth data with no injected notch, $|z|$ must stay under 3. Together these are the closest thing to a proof the implementation is correct: it finds what is there and stays quiet when nothing is.

Reproducibility: the cleaned panel is committed so notebooks run offline; raw archives are gitignored; seeds are fixed (42 for the ticker draw); versions pinned against the environment the analysis actually ran in.

Also produced: publication figures in PNG/SVG/vector PDF, a 4-page writeup, a 19-slide deck with speaker notes, an Excel workbook where histograms are live `COUNTIFS` over the panel, and a Google Sheets edition with the statistics as live formulas verified to match Python to three decimals.

PART 9 – ANTICIPATED QUESTIONS

25. Questions and answers

Written to be said out loud.

The result

"So did you find anything?"

Not a significant result, and that is the honest headline. The direction matched at every bin width – fewer small losses, more small profits – but the deficit below zero was never significant and the surplus above was borderline, around $p = 0.05$. What I did find is more interesting: the result depends heavily on what you divide profit by, and a placebo test on

cash flow showed the method itself producing false breaks. So my conclusion is a measurement one rather than a behavioural one.

"Isn't a null result a failure?"

It would be if I had designed the project to only work if the answer was yes. I planned for the null from the start, because the alternative is being tempted to torture the data. What I ended up with is a complete measurement study: here is the distribution, here is how it responds to every choice, here is what would have to be true for the classical result to hold. And the two findings that came out of it — denominator dependence and the placebo failure — are more useful than a marginal confirmation would have been.

"Why didn't you find it when the 1997 paper did?"

Three candidates and I cannot fully distinguish them. Period: they used 1976–1994, I used 2016–2025, and the notch may have faded after Sarbanes-Oxley raised the cost. Method: my placebo suggests the smoothness test is unreliable on this data, which raises a question about the original too. Sample: XBRL-era filers are a different population from the Compustat universe of the 1980s. The time-trend split is the check I would run first, but SEC structured data does not reach before 2009.

"Which finding are you most confident in?"

The denominator sensitivity. It is a within-sample comparison — same companies, same years, same code, only the divisor changes — so it is not confounded by period or sample construction the way a comparison to a 1997 paper is. And it reproduces at all three widths. The caveat I add unprompted is that the revenue sample is not quite the same set of companies.

"Your surplus above zero is $p = 0.051$. Isn't that basically significant?"

It is basically *nothing*, and that is the point. A result at 0.051 and one at 0.049 are the same evidence; only one gets a label. Treating 0.051 as a discovery is the borderline reasoning that fills literatures with findings that do not replicate. What makes me comfortable calling it marginal is the context: the other side of zero shows nothing, and the placebo shows the method generating false positives.

The method

"Explain the test to someone non-technical."

Line every company up by profit as a percentage of its size and count how many fall in each narrow band. For any band ask: if the pattern were smooth, how many should be here? The natural answer is the average of the two bands either side. Compare that to what is actually there. If a band is much emptier than its neighbours suggest, something unusual is happening. The z-score measures "much" in units of the random variation you would expect anyway.

"Why three bin widths?"

Because bin width changes the answer, and reporting one invites "did you pick the flattering one?" Too wide smooths the notch away, too narrow makes every bin noise. It also turned out diagnostic: the cash-flow effect *grew* with bin width, which is the signature of curvature rather than a genuine discontinuity. I would never have spotted that from a single width.

"Why must zero be on a bin edge?"

Because the hypothesis is about a break at exactly zero. A bin from -0.0025 to $+0.0025$ would contain both the missing losses and the extra profits, which average each other out. I would be deleting the signal before measuring it. My edges are integer multiples of the width, and a test fails if that changes.

"Where does the standard error come from?"

Each observation either lands in a bin or not, so counts are binomial with variance $N \cdot p \cdot (1-p)$. The prediction is built from two neighbouring counts which are also random, so their variance adds — and because I use their average rather

than sum, that term carries a factor of one quarter. Add the two and take the square root. I also ran a second test that does not depend on the neighbour assumption: fit a smooth curve with the two zero-adjacent bins held out, then test those bins against it.

"How do you know your test works?"

I tested it where I know the answer. I take smooth random data, move half the bin below zero into the bin above — injecting a notch of known size — and assert strong detection in both directions. Then the mirror: on smooth data with no notch it must not fire. Both are unit tests that run on every change.

"What would have convinced you the effect was real?"

A deficit below zero significant at all three widths, surviving all three denominators, and absent from the placebo. I got none of the three. I would also have wanted the effect not to grow monotonically with bin width.

The data

"Why SEC data rather than a commercial database?"

Free, primary, and reproducible by anyone without a licence — which matters for a project someone might check. The trade-off is coverage starting in 2009, XBRL-era filers only, which is a real limitation and one candidate explanation for my null.

"How did you handle the volume?"

113 million rows across 37 archives. I parse the numbers file by hand rather than with `read_csv`, because the files are ~550 MB each with a free-text footnote field that breaks naive quoting. Splitting on tabs and filtering to six tags before materialising anything keeps memory flat and lets me count malformed lines exactly rather than silently skipping them. Zero malformed across all 37 quarters — a verified fact, not an assumption.

"Why 10-Ks only? Didn't that throw away data?"

It threw away quarterly filings, but bought the thing I most needed. A 10-K shows last year's balance sheet beside this year's, so one filing gives both the profit and the beginning-of-year assets I divide by, from the same audited document. No cross-filing join, no risk of mismatching a company to the wrong prior year. Given the lagged denominator is the most important measurement decision, that was worth the trade.

"You dropped banks — a quarter of your data."

Because total assets does not mean the same thing for them. A bank's assets are its loan book; an insurer's its investment portfolio. Both are an order of magnitude larger relative to earnings than a manufacturer's factories, so ROA is not comparable across that boundary. Including them means the distribution is a mixture of two populations with different natural scales, and a kink at the join would be my own construction. Standard in the literature, and in the drop log with a count so nobody takes my word for it.

"What about survivorship bias?"

Real, and I cannot measure it. Only companies filing with XBRL tags between 2017 and 2026 appear, so anything that went bankrupt or deregistered before 2017 never does. Loss-makers are likelier to disappear, so the region just below zero is probably affected more than above, which would push toward an apparent notch. I flag it rather than pretending the sample is complete.

The critique

"What's the Durtschi–Easton critique in one sentence?"

That a discontinuity in scaled earnings can be produced by the scaling and by sample selection rather than by managers, so the shape of the distribution is not evidence of manipulation by itself.

"How did your design respond to it?"

Four ways, mapping onto the four robustness checks. Alternate denominators test the scaling directly. Size terciles test their specific prediction that the effect concentrates in small firms. Size-floor sensitivity tests whether my own cutoff does work. The placebo tests whether whatever produces the break could plausibly be earnings management at all. Three of four came back pointing away from a management interpretation.

"Did your results support the critique?"

Partly, and not as I expected. The denominator sensitivity supports it directly. But the size checks cut *against* the specific small-firm mechanism they emphasise. And the placebo found something neither paper predicts: a significant opposite-signed break in cash flow, suggesting the smoothness assumption itself is unreliable, independent of scaling.

"If the method is unreliable, does that invalidate the 1997 paper?"

I would not go that far. What I showed is that this method produces a significant break in a control variable on *my* data, in *my* period, with *my* filters. That is grounds for caution about results resting on the same assumption, including mine. It is not a demonstration the original was wrong — different period, different sample, much larger effect. The honest statement is that the null deserves more scrutiny than it usually gets, and I would want the placebo run on their data before saying anything stronger.

Judgment and process

"What was the hardest decision?"

Which denominator to make the headline. Revenue gives the significant, publishable-looking result. Assets is the convention, what the original paper used, and what I had committed to before seeing output. I went with assets and reported revenue as a robustness finding. Flipping that after seeing results would have made the project p-hacking with extra steps — and what makes the denominator finding interesting is precisely that I did not choose the flattering one.

"What would you do differently?"

Model the density properly — the neighbour-averaging null is the weak link and my own placebo showed it. I would fit a smooth model to the whole distribution and test a break against it, which would also give a confidence interval on the notch size rather than only a test against zero. Second, build a matched sample with all three denominators available, to separate the scalar effect from the sample effect. Third, try harder on market value of equity by joining company IDs to a price source.

"How long did it take, and in what order?"

Phases with a gate at each. A crude 300-ticker prototype first, to see whether the effect was visible before committing — not conclusive, but that was the point of doing it cheaply. Then the modules and unit tests. Then the SEC pipeline and documented sample. Then the main test at all three widths. Then robustness. Then the writeup and figures. The order matters: filters and guardrails were committed to code before any result existed.

"What are you most proud of?"

That it is honest under pressure. Every exclusion counted, every specification reported, the phrasing never exceeding the evidence — and when the placebo came back pointing the wrong way I wrote it up as the most interesting finding instead of burying it. The code enforces the discipline too: the zero-on-a-boundary rule and the lagged denominator are protected by tests rather than by my memory.

"Given a week, what else would you look at?"

The other threshold from the original paper — zero *change* in earnings year over year, the second place managers are said to avoid crossing. Same machinery, different variable, so it is cheap and a genuinely independent test. If the notch appeared there but not at zero profit, that would be interesting; if neither, that strengthens the null. Then the time trend, if I could find a source reaching past 2009.

"What did you learn?"

Three things. Decide your filters before you look — it is the difference between a study and a fishing trip. Plan for a null, so a clean "no" is still complete work. And build the placebo early: cash flow was one extra column of data and it changed how I read everything else.

Cheat sheet

Pipeline — 37 quarterly SEC archives (2017 Q1 – 2026 Q1) · 113,203,537 fact rows scanned, 0 malformed · 244,697 submissions, 53,156 annual 10-Ks used · ~3,200 lines of Python · 35 tests passing

Sample — 57,407 raw firm-years → **31,240** after filters, **5,366** companies · **16,018** firm-years from **3,582** companies in the $\pm 10\%$ window · FY2016–2025 · 5.82 years per company · median lagged assets \$615m · 50.0% of firm-years report a loss · largest exclusion: financial firms, 14,590

Main test — net income ÷ lagged assets

Width	Below zero	z	p	Above zero	z	p
0.0025	246 vs 252	−0.31	0.75	291 vs 253	+1.91	0.057
0.005	459 vs 480	−0.79	0.43	550 vs 496	+1.95	0.051
0.01	868 vs 899	−0.87	0.38	1,083 vs 1,017	+1.73	0.083

Held-out χ^2 p: 0.0009 / 0.0052 / 0.0044 — driven by the surplus above zero.

Robustness at width 0.005

Check	z below	z above	Verdict
Revenue scaling	−2.97 (p .003)	+4.62 (p <.001)	significant at all widths
Book equity scaling	−1.20	+1.61	not significant
Size terciles (S/M/L)	−0.25 / −0.26 / −0.91	—	no small-firm concentration
Size floors 0 → 500m	−0.72 to −1.09	+1.46 to +2.08	flat across 50× range
CFO placebo	+2.04 (p .042)	−1.79	opposite sign
CFO placebo, width 0.01	+3.55 (p <.001)	−3.59 (p <.001)	grows with width

Measurement — lagged vs end-of-year assets identical in 0.02% of firm-years · median absolute difference in ROA 0.0093 (≈ 2 bins at width 0.005)

Citations, verified against publishers — Burgstahler & Dichev (1997), *J. Accounting and Economics* 24(1), 99–126 · Durtschi & Easton (2005), *J. Accounting Research* 43(4), 557–592

Glossary

Accrual — entry recording revenue/expense before the cash moves. **Accrual accounting** — recording when earned/incurred rather than when cash moves; required for public companies. **Audited** — examined by an independent firm that issued an opinion. **Balance sheet** — what a company owns and owes at one instant. **Bin** — one histogram interval, [left, right). **Binomial distribution** — count of successes in N yes/no trials; variance $N \cdot p \cdot (1-p)$.

Book equity — assets minus liabilities. **Cash flow from operations (CFO)** — cash the business generated; contains no accruals. **Chi-square** — goodness-of-fit statistic summing $(\text{obs} - \text{exp})^2 / \text{exp}$. **CIK** — the SEC's permanent company identifier. **Debt covenant** — a loan condition, e.g. staying profitable; breaching one can trigger repayment. **Degrees of freedom** — independent pieces of information in a test. **Discontinuity / notch** — a sharp break in a smooth

distribution. **Distribution** — how often each value occurs across a group. **Earnings management** — steering reported results using discretion within the rules; legal. **EDGAR** — the SEC's public filings database. **Firm-year** — one company, one fiscal year; the unit of analysis. **Fiscal year** — a company's own 12-month reporting period. **Goodness-of-fit test** — compares observed counts to a model across many bins. **Histogram** — a chart of a distribution. **Income statement** — what a company earned over a period. **Lagged** — from the previous period; lagged assets = last year's closing balance sheet. **Liabilities** — what a company owes. **Net income** — bottom-line profit after all expenses and tax. **Null hypothesis (H₀)** — the "nothing unusual" default. **Panel** — one row per company per year. **Parquet** — compressed, typed, column-oriented file format. **p-hacking** — trying many choices and keeping the most favourable. **Placebo test** — the same analysis on something that should show no effect. **Pooling** — treating every firm-year as independent. **p-value** — probability of a result this extreme if the null were true. **Revenue** — sales before costs. **ROA** — net income ÷ total assets (here, always lagged). **ROE** — net income ÷ equity. **Sampling variation** — wobble from a limited number of observations. **Sarbanes-Oxley (2002)** — post-Enron law requiring executive certification of financial statements. **Scaling** — dividing by a size measure for comparability. **Shareholders' equity** — assets minus liabilities. **SIC code** — four-digit industry classification. **Significance level** — the threshold for calling a result significant, conventionally 0.05. **Standard error** — typical variation of a statistic from randomness alone. **Statistical power** — probability of detecting a real effect. **Survivorship bias** — distortion from studying only what survived. **Tercile** — one of three equal groups. **Total assets** — everything a company owns of value. **Type I / Type II error** — false positive / false negative. **Variance** — square of the standard deviation; variances of independent quantities add. **XBRL** — tagging standard making filings machine-readable. **z-score** — standard errors from the expected value; $|z| > 1.96$ is the 5% threshold.